MILITARY SPECIFICATION

AMPLIFIER, ELECTRONIC CONTROL, SYNCHRO SIGNAL AM-6015/A

1. SCOPE

1.1 Scope. This specification covers the requirements for an electronic control synchro signal amplifier designated as AM-6015/A. The amplifier is used to provide increased torque to synchro torque receivers and minimize interaction between receiver loads, and as an isolation amplifier for control transformer loads.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-20708/5	Synchro Torque Receiver, Type 26V-11TR4C
MIL-S-20708/7	Synchro Control Transformer, Type 26V-11CT4D

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (System and Equipment)
MIL-STD-831	Test Report, Preparation of
MS3102	Connector, Receptacle, Electric, Box Mounting

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6615

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3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-E-5400 shall apply as requirements of this specification with additions or exceptions to the paragraphs of MIL-E-5400, as identified by similarity of paragraph headings, called out herein. When the two documents conflict, this specification shall govern.

3.2.1 Classification. The AM-6015/A amplifier shall be class 2 as defined in MIL-E-5400 with exceptions as specified herein.

3.2.2 Materials, parts, and processes

3.2.2.1 Approval of nonstandard parts and materials. Contract requirements for nonstandard parts and materials approval shall be category III as defined in MIL-E-5400.

3.2.2.2 Electrical connectors. The electrical receptacle for the amplifier shall be type MS3102R18-1P and shall be designated J-20.

3.2.2.3 Marking and identification

3.2.2.3.1 Label. The information specified in 3.3.2 shall appear on a label attached to the amplifier.

3.2.2.4 Electron tubes and semiconductor devices. Electron tubes shall not be used in the design of the amplifier.

3.2.3 Design and construction

3.2.3.1 Interference. The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-I-6181.

3.2.3.2 Reliability, and maintainability

3.2.3.2.1 Reliability program. The contractor shall establish and conduct a reliability assurance program in accordance with MIL-STD-785.

3.2.3.2.2 Reliability. Reliability of the AM-6015/A amplifier shall be determined in accordance with MIL-STD-781. The amplifier shall have a minimum acceptable mean-time-between-failures (MTBF) of 5000 hours during its useful operating life.

3.2.3.2.3 Maintainability design requirements. Maintainability design requirements shall be demonstrated in accordance with MIL-STD-471.

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3.2.3.3 Electrical overload protection. The amplifier shall withstand a short circuit to ground on either or both amplifier outputs for not less than 10 minutes. Upon removal of the short, the amplifier shall resume normal operation without resetting or replacing a fuse.

3.2.3.4 Service conditions (electrical). The amplifier shall be designated as a category B utilization equipment in accordance with MIL-STD-704.

3.2.3.4.1 Total input power. The required 115V, single phase, a-c power shall not exceed 300 va.

3.2.3.4.2 Warmup time. The amplifier shall not require warmup time.

3.2.3.4.3 Repeater excitation. The amplifier shall supply 26V, 400 Hz ac at 1 ampere maximum for use as torque receiver excitation. All torque receivers driven from the amplifier shall be excited from either the 115V, 400 Hz input power or the 26V, 400 Hz excitation voltage.

3.2.3.5 Service conditions (environmental. Class 2 equipment requirements of MIL-E-5400 shall apply except as specified herein.

3.2.3.5.1 Temperature. The intermittent operation of the equipment shall be at 95°C.

3.3 Design and construction

3.3.1 Form factor. The form factor of the Amplifier, Electronic Control, Synchro Signal AM-6015/A shall be in accordance with figure 1.

3.3.2 Receptacle pin designations. The J-20 amplifier receptacle shall have the pin designations as shown on table I.

3.4 Performance

3.4.1 Impedance

3.4.1.1 Input impedance. The input impedance shall be 10,000 ohms minimum.

3.4.1.2 Output impedance. The output impedance shall be 2 ohms maximum when working into an 100-ohm load with an output voltage of 10V rms at 400 Hz.

3.4.2 Load capability. The amplifier shall be capable of positioning four parallel connected torque receivers, type 26V-11-TR4C (MIL-S-20708/5B), or drive four control transformers, type 26V-11-CT4D (MIL-S-20708/7B).

3.4.3 Accuracy. The amplifier shall not introduce more than 0.1 degree of error into a torque receiver load.

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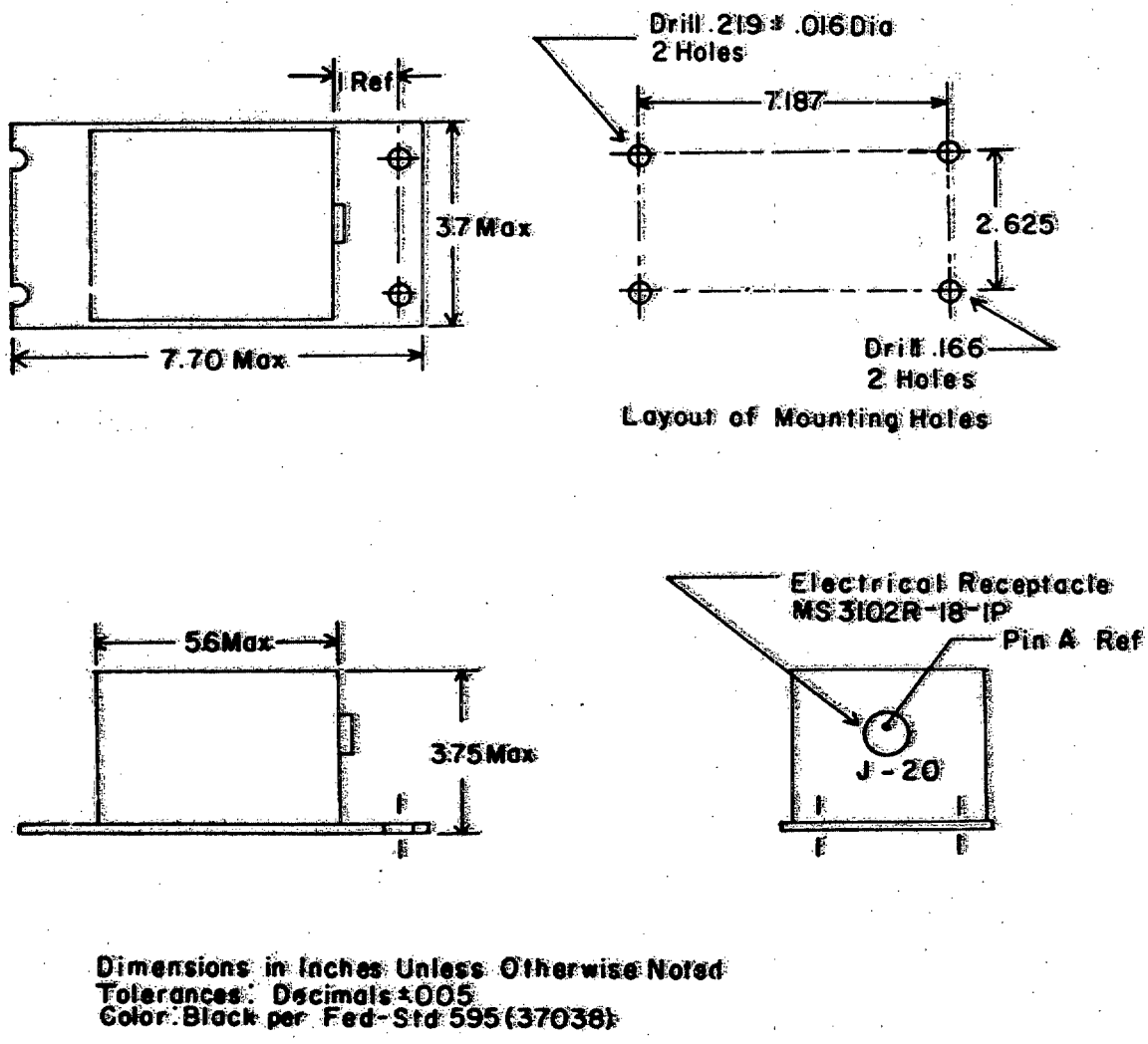
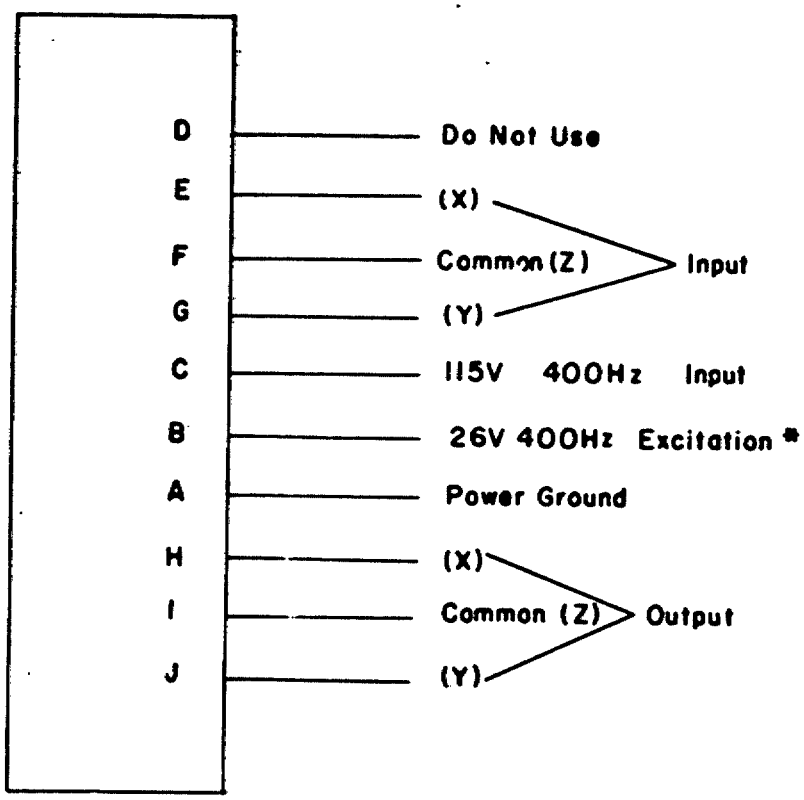


FIGURE 1. Outline of Amplifier, Electronic Control, Synchro Signal AM-6015/A

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TABLE I. Pin Designations of J-20 Receptacle



* Note: 86.5V 400Hz may be obtained on pin B by internal wiring modification.

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3.4.4 Multiple load isolation. The displacement of one of the multiple loads 20 degrees from the transmitter output shall not cause any other receiver load or transmitter to vary more than 0.2 degree.

3.5 Weight. The weight of the AM-6015/A amplifier shall not exceed 3 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The quality assurance provisions of MIL-E-5400 and the requirements specified herein shall be applicable to this specification. When the two documents conflict, this specification shall govern.

4.2.1 Classification of inspections. The examining and testing of the amplifier shall be classified as follows:

a. Preproduction inspections

b. Quality conformance inspections

4.3 Test conditions. Unless otherwise specified, the amplifier shall be subjected to the tests under the following standard test conditions:

Temperature:	Room ambient (25°C ±5°C)
Altitude:	Normal ground
Vibration:	None
Humidity:	Room ambient up to 90 percent relative humidity
Input voltage:	115V ±1.0V, 400 Hz, single phase

4.4 Preproduction inspections

4.4.1 Preproduction samples. Preproduction samples shall be representative of the production amplifier. They shall be tested under the conditions specified herein and at the point designated by the contract.

4.4.1.1 Testing instructions. The supplier shall conduct tests on the preproduction samples to determine that the design of the equipment meets the requirements specified herein. The supplier's tests shall be conducted in accordance with the test requirements specified herein.

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4.4.1.1.1 One preproduction amplifier shall be subjected to all the preproduction tests specified herein.

4.4.1.1.2 Preproduction reliability testing shall be in accordance with test level F of MIL-STD-781. Quantities to be subjected to the preproduction reliability test and accept-reject criteria shall be in accordance with test plan XXVI of MIL-STD-781.

4.4.2 Preproduction tests. Preproduction tests shall consist of the tests specified under 4.6.

4.5 Quality conformance inspections. Quality conformance inspections shall consist of individual tests and sampling plans and tests.

4.5.1 Individual tests. Each amplifier shall be subjected to the following tests:

Tests	Applicable paragraph
Examination of product	4.6.1
Burn-in	4.6.2
Power consumption	4.6.3
Voltage gain	4.6.4
Accuracy	4.6.5
Multiple load isolation	4.6.6
Overload	4.6.7
Repeater excitation voltage	4.6.8

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. The reliability production acceptance test shall be conducted in accordance with the requirements of test level F of MIL-STD-781. The quantity to be tested shall be as specified in table II.

TABLE II. Sampling plan A and B quantity selection

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.5.2.1.1 Accept-reject criteria. Accept-reject criteria shall be in accordance with test plan XXVII of MIL-STD-781.

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4.5.2.2 Sampling plan B. Amplifiers selected at random in accordance with table II, and having passed the individual tests, shall be subjected to the following tests:

Tests	Applicable paragraph
Overload test (The amplifier shall operate with a short for 10 minutes.)	4.6.7
Temperature and altitude	4.6.9.2
Vibration	4.6.9.3

4.5.2.2.1 Rejection and retest. When one amplifier selected from the production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The supplier shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.3 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the amplifiers on hand or later produced shall not be made until it is determined that all amplifiers meet all the requirements of this specification.

4.5.4 Defects in amplifiers already accepted. The investigation of a test failure could indicate that defects may exist in amplifiers already accepted. If so, the supplier shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The amplifier shall be inspected to verify that the materials, design and construction, necessary mechanical measurements, marking, and workmanship comply with this specification.

4.6.2 Burn-in test. Each amplifier shall be operated under standard operating conditions at a maximum applied voltage of 118V ±1V, single phase and minimum applied frequency of 380 ±5 Hz for a period of 10 hours.

4.6.3 Power consumption test. The total power required for the amplifier shall not exceed 300 va starting and 100 va running. For the above measurement the amplifier shall be excited by an 11.8V transmitter (or equivalent) and loaded with four torque receivers, type 26V-11-TR4C (or equivalent).

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4.6.4 Voltage gain test. The amplifier shall be delta connected to a 100-ohm load. With an input voltage of 10V rms, 400 Hz, the voltage gain of the two channels shall not differ by more than 0.5 percent.

4.6.5 Accuracy test. The amplifier shall be connected with a load of four torque receivers, type 26V-11-TR4C (or equivalent). With an input signal from a transmitter of known accuracy, the output data shall be measured every 30 degrees. The maximum error between transmitter data and the amplifier output shall be less than 0.2 degree.

4.6.6 Multiple load isolation test. The amplifier shall be connected with a load of four torque receivers, type 26V-11-TR4C (or equivalent). One of the torque receivers shall be displaced 20 degrees. The heading output of the amplifier shall not deviate from the transmitter data by more than 0.3 degree.

4.6.7 Overload test. The amplifier shall be connected with a data transmitter (or equivalent) input. Both outputs shall be shorted to ground and the amplifier allowed to operate for five minutes. Immediately following, the amplifier shall be connected as specified in 4.6.5 and rechecked for accuracy as specified in 4.6.5, except the measurement shall be every 90 degrees. The same accuracy tolerance shall apply.

4.6.8 Repeater excitation voltage test. The torque receiver excitation voltage shall be measured across a 26-ohm 50-watt resistor. The voltage shall be 26V ±1V.

4.6.9 Environmental tests. Environmental tests shall be conducted in accordance with class 2 requirements of MIL-T-5422 with exceptions or additions as specified herein. When the two documents conflict, the requirements of this specification shall govern.

4.6.9.1 After completion of each of the following tests, the amplifier shall meet the requirements of 4.6.5 and 4.6.6, unless otherwise specified:

a. Temperature and altitude: The tolerance specified in 4.6.5 shall be 0.3 degree; the tolerance specified in 4.6.6 shall be 0 35 degree.

b. Vibration, part I, curve I

c. Shock

d. Humidity

e. Salt spray

f. Explosion, procedure I

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g. Sand and dust

h. Fungus resistance

4.6.9.2 Temperature and altitude test

a. Step 1. The temperature of -62°C shall be maintained for 4 hours.

b. Step 5. The temperature shall be 85°C and shall be maintained for 4 hours.

c. Steps 7 through 13 shall not be required.

4.6.9.3 Vibration test. Step 6 of part I shall not be required.

4.6.10 Interference test. The amplifier shall be subjected to interference tests in accordance with MIL-I-6181.

4.6.11 Maintainability demonstration. Maintainability demonstration shall be conducted in accordance with test method 5 of MIL-STD-471.

4.7 Design data and material

4.7.1 Preproduction test report. The preproduction test report shall be prepared in accordance with MIL-STD-831.

5. PREPARATION FOR DELIVERY

5.1 The AM-6015/A amplifier shall be prepared for delivery in accordance with the instructions of the procuring activity.

5.2 Reinspection markings. Reinspection markings shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The Amplifier, Electronic Control, Synchro Signal AM-6015/A covered by this specification is intended to provide increased torque signals from the compass gyro standard heading transmitter to standard receiver (indicator, autopilot, navigation computer, and radar set).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Stock number

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c. Preproduction samples and tests

(1) Number of preproduction samples

(2) Point of inspection

(3) Requirement for concurrent delivery of each sample and its test data. The procuring activity should be given at least 5 days prior notice when the preproduction tests are to be conducted so that they may be supervised or witnessed, if desired, by a Government representative.

d. Submission of test programs for approval (preproduction, acceptance, and reliability)

e. Selection of applicable levels of preservation and packaging, and packing.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 6615-F004

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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